

ActInf Textbook Group ~ Cohort 4 ~ Meeting 3 (Chapter 1 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 3 (Chapter 1 part 1) | 56:59

all right greetings everyone
welcome to cohort 4 on June 20th for
chapter one discussion
we're going to get to chapter one
discussion in one second but I wanted to
just show one area of development and
something that people can explore
um and then mention one thing about math
learning group so first in the equations
page
there's a few sub pages that might be
interesting to explore
first where we previously had the
equation screenshots so you could like
Mouse over equation 2.1 and click on the
screenshot
um we added the latex so this is just a
plain text code that renders in coda
then there's some sub views that use
Coda AI so this explanations View
these prompts
um basically just to show what it looks
like in code AI explain the latex as if
I'm not familiar how would a high
schooler be interested in this what's
some summary code where is this equation
important and can you explain the
variables it's not perfect but it gets
pretty far very cool
this is the explanations page there's a
philosophy page just one column but it's
just bring up 10 philosophy questions
and a lot of these are like awesome
discussion starting points
you know what what does Divergence mean

all these things and then also languages
so here's just a few languages
but we can add
people's preferred languages just
totally feel free to make a suggestion
we can add more AI columns
um so we have to keep on exploring this
because the latex representation with
plain text is really amenable to this
kind of language model work
so just mention that and then also
um someone in cohort for is interested
in the math learning group I think
they're

[Music]

um
they're potentially interested in like a
morning time UTC
but if somebody else is interested then
we can kind of get this going
so it could be in the Discord
but if anyone is interested to pick up
with math learning group and basically
just it could be an unrecorded session
in the Discord

another hour

um to go over the equations

um

sorry about that um

so if anyone is interested

um maybe just make write a comment here

and just say here sometimes I'd want to

do this just basically like a math study

time

um and then let's just see if there's a

any specific options for what times

people want to do such activities so

on either the AI or the math learning

group does anyone have any just general thoughts or questions otherwise we'll move to chapter one

all right

so

let's head over to chapter one

um

just right off the bat before we get to the

um

questions

it would be awesome if anyone could just share any of their Reflections on chapter one just anything about their experience reading it or what what they felt it was like or what stuck with them or anything that kind of raised after they

um

read chapter one

um yeah

um so I've had some familiarity with active inference for a while so a lot of what I read already tracks for me

um but that said

I'm curious about variational uh free energy is like a particular

it seems that so active inference seems to be based on a particular form of Bayesian inference that it relies upon like this variational formulation I'm more curious about that

I assume the textbook will get much more into

what's specific to variational free

energy is as opposed to some other
um formulation of it but yeah that's
something that that I'm kind of still
chewing on right now
thank you
anyone else
Yeah I'd write a chapter one a while ago
and
I think I just Dove deeper into it
knowing that you know we're looking for
questions and stuff but I really
appreciated the um the video that you
and Ollie did so I watched that for the
first time
today and I really appreciated the heads
up to actually read ahead to chapter 10
which I'll do yeah that looks good
um for me in this this more in-depth
reading um action really stuck out to me
um so I went to the ontology to look at
the definition of action
that's like Central to my area I'm in
motor neuroscience
uh
and I can relate to the idea that action
has to have a goal of purpose but I
noted that in the ontology that goal
purpose is not in the definitions
it's more of a general sort of
Behavioral definition
um
and then I started to think about you
know in our area we also use
inaction so like action imagery
and observational learning which you're
not really acting you're watching
somebody else
and and so I posted that question about

um does Action have to change the world
and what do we mean by that do we mean
like I'm going to change the physical
world that we all share
or am I actually sharing my internal uh
changing my internal model
of the world through action
yeah so in summary I really got hung up
on this this action
um terminology
cool that that those are some great
things to explore
anyone else with a just any thought
about chapter one
and
I I'm from I was familiar with the
Bayesian brain hypothesis about like the
you know our mind is kind of uh a
predictive mechanism and
yeah but but like this framework
um you know the more I thought about it
the more I started seeing like these
patterns around me like thinking about
you know the different uh things that I
observe in in that sense of active
inference like uh the action and the
perception
and and like specifically I saw it like
in children's Behavior
for example like uh
you know that the ability to to change
their mind and and the relief
in contrast with like stubbornness
sometimes you know when they're very
stubborn and kind of like a you know
they will do whatever they can to change
their reality although you know they
can't necessarily do so but so it was it

was very interesting just uh thinking
about like uh various behavioral
um patterns that I see around me and
translate them into this framework
thank you

yeah of here and then anyone else uh
question about uh salency uh because I'm
quite familiar with the definition of
salinity in vision science in which
something is silent if you know the
visual features involved are different
from what you have in your neighbor
but I think that silencio has a much
more general definition in active
inference some quite interesting to
understand you know how it is Quantified
and what the Yeah it's proper definition
you know it's generalization how it
generalize you know the very simple
definition in uh in vision science
yeah

cool uh for me uh I'm a psychiatrist so
um talking about Helen Hall uh Freud who
invented the cycle analytic theory was
influenced by holy molds and as you can
see in the chapter one and also at
chapter two uh Helen Holmes said that
perception is an unconscious inference
process
and which is quite interesting because
um

so the cycle analytic theories are based
on uh the assumption that individuals uh
make really individual uh
interrelationship based on their
childhood memories and experiences and
based on the fact uh I I I got quite
curious about uh whether we can explain

psychoanalytic theories based on active inference or other computational models that are based on these assumptions or theories that are inspired by Helen holds or other uh

other and other scientists who are also having similar similar uh points cool

there's definitely some other people in the in the textbook group and around so I hope that they can continue on this topic anyone else want like a general

um topic or we can return to some of these mentioned points as well as some of the other asked questions

um

just uh I don't want to track too much

but just in response to Young Boon uh I

and maybe you're already familiar with

this but the way I ended up in active inference uh was through the work of

Mark songs uh and neuropsychanalysis I

yeah so I I would strongly recommend

checking him out and he he wrote a piece

where he completely wrote rewrites

Freud's uh like early scientific paper

um using the terminology of of active inference

um I think it's great and I'd be happy

to explore the relationship between

psychoanalysis and active inference like

more potentially within the confines of

of this group in the textbook

oh yeah I I heard about him but uh I

never actually uh read his papers so

I'll look it up thanks thanks for

bringing up

sure

thank you to Daniel

well you want to unmute and go for it or

he I just wanted to mention who worked

North half's work in relation to

um I mean psychoanalytic understanding

of active inference uh it's not

completely in line with Mark Stones line

of research but it provides an

alternative path uh through some of the

uh psychoanalytic ideas of active

inference based on his own spatial

temporal philosophy in Europe

okay cool

all right well let's just revisit some

of these topics and then turn to the

questions but we're totally

just killing and and whichever

um ways that people want to go or anyone

who wants to raise your hand like you

know what

whatever comes to mind let's just

do them while we're here

um

some of these

as with the many coats of paints idea

we're just gonna kind of give like a

first

or a few thoughts on and we're going to

be revisiting in the coming chapters

because like this is the topic chapter

one is just kind of like opening the

door

so let's we're gonna certainly address

all of this once

all these topics are going to come back

like again and again

because the ontology gets Revisited so

much so

um

active inference being based upon the
variational free energy as opposed to
some other formulation

we're going to talk about this a lot
more in chapter two and in equation 2.5
we're going to meet the variational free
energy

and so leading up to equation 2.5 and in
this discussion around 2.5 and in
Chapter 2 and elsewhere

we're going to talk about like what do
we mean by variational free energy and
why is active inference using it
essentially it's a really tractable and
optimizable heuristic

that lets us carry out Bayesian
inference efficiently because there's
situations where if you had infinite
Computing resource you could do exact
Bays you could just plug and chug and
just do the matrix multiplication and
get the exact correct answer

however with large data sets that's not
really tractable and so in Bayesian
statistics and this is even outside and
before active inference people use
variational methods because they're
super effective computationally so for
example a variational autoencoder
a variational autoencoder is a widely
used machine learning method or a
Bayesian statistical method
and a variational autoencoder finds the
optimal fit for data not underfitting or
overfitting

through finding the variational free

energy

so inactive inference a lot of times
there's there's a lot of interpretation
and a lot of extension of what free
energy is in all of this but it's also
possible to kind of pull back to this
pretty non-contentious

Bayesian statistical interpretation
and just

consider variational methods
as being part of Bayesian statistics and
used to tractively optimize challenging
inference problems

so we're going to talk more about that
in chapter two anyone have a thought or
a question on this

um Ali please

I wanted to

point to the distinction between
variation of pre-energy and expected
free energy because

as it says here active in prints is
based upon the variation formulation
it's true but uh

the perceptual part of active inference
theory is based on variation of free
energy but when it comes to action
policy selection and decision making the
parameter to be to be optimized is
actually expected free energy

which is expressed in equation 2.6 so
basically those two equations between
variationally free energy expressed in
equation 2.5 and the expected free
energy expressed in equation 2.6 can be
seen as parallel to each other but uh
they're used to formulate different
aspects of uh for of active inference

one is for perception and the other one
is for action

awesome thank you Ali two ways to
minimize free energy
update your mind through perception and
learning

action through changing the world
the only thing you can't do is like put
your thumb on the scale and directly
change your sensory outcomes so you
prevent the thumb going on the scale
because you don't have a line going from
the internal States out to the sense
States

so that the set states are only coming
in that's inbound dependency
and then that leaves two avenues for
reducing
um free energy perception and action and
we're going to talk a lot more about
this in chapter two when equation 2.5
and 2.6 come into play

okay
continuing on this theme of action
so action was brought up and obviously
like this is one of the really exciting
pieces

this is what is going to be referenced
throughout the book
a unified approach on perception and
action or all these different ways which
we see active inference presented
like going Beyond passive or perceptual
inference

so does Action have to change the world
is this changing the physical
environment or can we change the
generative model through action

thank you um Holly

uh yes so uh maybe I'll just uh yeah

please I think you know I think you know

the when Ali brought up expected free

energy you know you're you're making

changes to your policy your brain uh

your model and

um and so then what's happening is that

and what I find interesting uh is the

leverage points and this concept that

making a very slight change in your

understanding of the world has a much

larger effect than an actual you know

physical action you can see it um there

in that image so I thought and I don't

know if I don't know if a variation of

the energy and the expected for energy

really take that into account that you

know that these you know the The

Leverage points or not that maybe some

people have some ideas

well it's like like in nested modeling

like for a generative model that only

sees up to this point

the optimal leverage Point that's

detected is going to be so and so but

then a deeper generative model might be

able to see uh more leverage points

Ollie

uh well rather than describing action as

something that changes the environment

if we want to talk a bit more precisely

uh what action means in active inference

is actually

one in which active States intervene

constantly in external Dynamics right

and uh formally it means a kind of

Jacobian coupling between the external

states that depend upon the act upon the
act of the states or active intervention
so uh what we mean by changing the
environment uh we mean more precisely
the the kind of active state that would
lead to a kind of causal intervention uh
in the environment and not something
that just

I don't know it changes something per se
and this kind of this station comes into
play when we want to talk about uh
covert and over actions as discussed in
phenomenology which will come to later
in the book but yeah I think that would
be helpful to

awesome you have so many things to look
ahead to overt action is physically
manifested embodied action
covert action or internal action is
attention or mental action
and so that is like something that
doesn't necessarily

um
change outside of the generative model
but it's it's modeled with attention and
action inside the model and there's
actually

um even a very cool

um recent work on transparent and
explainable artificial intelligence and
again just looking ahead

here they use

attention and meta awareness as internal
or covert attention

uh generative models in an artificial
intelligence way but this model was
actually developed in a meditation and
phenomenology setting

where the action selection was internal
attentional control not external control
um

yeah I'll put the um this is the
explainable AI paper in the chat and
then also just again looking ahead one
place to to what is this concept of
action this is figure 4.3 which is kind
of like a Rosetta Stone figure
Pi are going to be the actions that are
selected from

and where those actions intervene is how
the hidden states of the world change
Through Time

so if the hidden States s changing
through time is like the true
temperature in the room
and at each time Point T you have a
thermometer reading that gets emitted
from the true temperature so we're only
able to observe the observables which is
the thermometer readings
and there's the true underlying changes
in temperature or you could think of
this as the true underlying activity of
the neural system and the observation is
the fmri that's kind of where this comes
from in terms of like friston's lineage
with SPM

and then

if there's two actions you can take turn
on the heater or not
it causally intervenes in how States
change their time maybe if you don't
turn on the heater it just passes the
hidden State on forward without changing
it and if you do turn on the heater it
increases it by one per time step

that's that's one sense of um action
and then also just by mousing over
um we see it from the ontology and also
we we can update the ontology it's it's
never like these things are locked
definitions just to be clear but we see
a really common pattern which is that
there's like a broad and a narrow sense
a lot of these terms in the active
anthontology action belief attention
salience

things that people have brought up today
there's like a broad sense qualitative
definition which largely or exactly is
how we would use this day to day
just action is what occurs
super general term and then there's like
a narrow or a more technical definition
which sometimes conveys how this is
operationalized in generative modeling
so more specifically
actions are the active States
that the agent's generative model
is going to select
and actually and so that's like more of
the quantitative and formal definition
of action

and then we'll explore a lot about how
his action selected
because that is the entire expected free
energy

imperative is the selection of action
and how goal orientation preference
orientation expectation orientation
is embedded Within action
despite there not being a reward
function as we might see in reward
learning reinforcement learning

we're going to have a different approach
for adaptive and flexible Behavior
that's not based upon reward learning or
reward function

okay any any other thoughts or just any
other ideas on on this in action Martin
so does Action have to have movement
it's one thing I'm looking at here I

mean the the overt versus the covert so
overall obviously I have to have some
movement that I could observe
um but he said uh earlier that you could
have the covert like for example
attention

being a covert action

yeah well depending on the system of
Interest action could be an API call so
certainly it doesn't have to be a
person's body that we're modeling could
be a computational or a purely
informational system

um

and then covert or internal attention I
mean you could argue it has some
physical informational substrate

so it's not like it's um you know
happening in some some you know
alternative plane of existence but for
the purposes of the modeling it might
not be realized through a gross
morphological movements

yeah and and it's going to be super cool
like how we're gonna when we look at um
figure 4.3 as kind of this cognitive
kernel or this kind of firmware element
and then again when we take it to uh
when we look at how these
um models compose

even just starting to see familiarity
here in figure two
we see basically this figure 4.3
downward facing e that's the inference
part
hidden State and observations and policy
intervening that's like this
downward facing e
policy intervening
and then that just gets nested
and so that speaks to the composability
of these kinds of Bayesian graphs
and then
in when you look at a specific model it
becomes clear which actions are
associated with overt movement and which
actions are associated with covert
attention
um
okay any other thoughts or thoughts or
questions
um yeah I guess just to briefly kind of
add to the table under
action
um like on page 10 ever so briefly
um the authors make the distinction
between
exploration and exploitation
and um that seems like a really
interesting or useful
division to me like so both of them seem
to be
both of them are related to minimizing
free energy or expected free energy in
this context and
um yeah exploration is to resolve
uncertainty so I view that as like
taking in new information versus

um exploitation which has to do more
with
yeah acting on the World to bring your
your observations into line with
whatever your your goal is uh or your or
your plan is your your set point
to use that the word from cybernetics
yeah

thank you and here's a really clear
claim in contrast with other Frameworks
policy selection so behavioral selection
inactive inference automatically
balances exploration and exploitation
now this is a um

very
strong version of the claim as we'll
explore it's not that simply making an
active inference model does
automatically balance it it's not like
just writing it out is going to make you
know the startup work and the ship work
and the plane fly and all of this so so
it can be read too strongly
it's not that it's automatically going
to be effective however it sets the
stage or it provides an accounting
system

for the the Adaptive finessing between
exploration and exploitation
and
um

we'll explore that more in chapter two
when we look at what is it that expected
free energy and variational free energy
are
and then how is it that that policies
are selected behaviors are selected that
are able to adaptively finesse between

more exploratory behavior when there's
information to learn and more
exploitative behavior when there's kind
of pragmatic value on the table and
maybe it's more important to just get
the pragmatic value rather than to um
continue learning
but that's where we see
um variation in
Behavior ultimately variation in
cognition and behavior when there's too
much attention paid to epistemic or to
pragmatic value
and that's a lot of where like these
like Precision Psychiatry cognitive
modeling things come into play
foreign
[Music]
yeah
any other thoughts or questions on this
it kind of connects to to even the next
point that was on
salience
so
um
salience
again just as it is in the ontology not
the it's the only or the final
definition but it's just the extent to
which a cue is going to be attended to
a cue that is not salient
has no attention to it
and something that's not being attended
to the consequences for the queue are
diminished in the limit case if you
didn't pay attention to something it
just you know
didn't cross your radar it doesn't

update the generative model
that's like attention to zero in The
Other Extreme case when the attention is
super high it's like you're updating
your prior to just whatever it is the
new data point is you perfectly attend
to it and it is like the is same as
having like no memory
so no attention is like
totally pushing your prior forward
full memory
full attention no memory
and then finding out what is an Adaptive
amount of remembering and learning and
forgetting

um

Salient observations are those that
resolve uncertainty
that's the information that we're
seeking out
is the information that reduces our
uncertainty about something
and then how much it reduces it
we're going to talk about some
formalisms especially in chapter two
for comparing like is this sequence of
observations or that sequence of
observations going to be more Salient
which one is going to have higher
epistemic value
which one is going to reduce my
uncertainty more
which one will I learn more by viewing
which one will update my prior is more
which one should I pay attention to more
those are all kind of like related
questions for like which information is
going to be the good information

and these are the the two sides of the
joint imperative

for expected free energy

Information Gain and pragmatic value

epistemic value

how good is the information

pragmatic value what is the utility

am I going to be at homeostasis

exploration exploitation

and so expected free energy is like this

unified imperative

that contains both of those elements

such that different policies can be

ranked or evaluated or upweighted

based upon how well they contribute to

both Information Gain and pragmatic

value

comes up in chapter two but just wanted

to kind of preview it

any other

um thoughts or questions or let's let's

go to the questions table and just like

the the questions that people are most

interested

okay first I'm going to go to the

chapter four or cohort four questions

so just to make sure that even if they

have a few upvotes that that all of your

questions can be addressed so we'll

first look at this question

and then we'll turn to chapter one

questions overall

so as always like you know whenever

you're watching or participating

even if it's a chapter that was long

discussed ago or you know up ahead

just add questions it's it's really

awesome and upvote them so that and

develop the discourse so we can have

good short answers for these

okay

page nine action minimizes free energy

and surprise by changing the world

okay so talked a little bit about that

previously

does this imply that the agent

implements action to alter their world

or their actions implemented to change

the Asians model of the world or both

can an Asian Implement action not solely

to minimize their free energy but free

energy in another agent for example

using action demonstration for

observational learning

yeah does anyone have any thoughts on

this

okay I'm just starting with the

observational learning two two quick

angles listen again everyone can add

more in live stream 53

um sequence

there were some paleoanthropologists

and we discussed observation learning a

lot

and they highlighted how a lot of times

people have focused on the kind of

moment of innovation for example in um

creating stone tools in in human

cultures in history

but it's not just about the one

innovator making the stone tool it's

actually in the moment of observational

learning

when the onlooker

is able to either understand or not

understand and ultimately adopt or not

adopt or to copy or not to copy that
behavior

and so they approached this from an
active inference perspective

and this also brings up ideas like
thinking through other Minds toom
which is kind of like the active version
of theory of mind

but thinking through other minds or
thinking with other Minds is more
explicitly like

one's generative model
in a social environment

also includes a generative model of what
the other
is thinking and doing
and so

um if if one is in an uncultured setting
where action demonstration can be used
for observational learning then these
are the kinds of dynamics that that are
possible to model so it's one angle is
yes it's been considered in a social and
cultured setting how like gesture and
demonstration and observation might play
a role in transfer learning

um

a more Technical and general response is
in multi-agent systems

you have

agent reducing their own free energy
making their own perception Action Cycle
but also you have like the joint
distribution you know you have the nest
mate going about their own way

but then in the aligned or the
healthiest or adoptive setting the
colony is also reducing its expected

free energy

so One agent or neuron or an Esme or a person

might undertake an action that increases their own surprise transiently as part of a distributed cognitive system

that has a joint free energy that is reduced through that action

so it's not necessarily like every single subunit is always and only reducing its free energy per se

but the Adaptive whole

is going to reduce its free energy and that might be

um entailing some subunit increasing its surprise

uh yes than anyone else

uh yeah I mean I think I'm I keep

thinking in a way as a psychiatrist so

uh it feels like like psychoanalytic

therapy itself is something a therapist

suing Trying to minimize the free energy of the client

because like they provide explanations and interpretations about why they are doing such Behavior or why they are making such relationships with others so

I guess maybe uh in a broad aspect I

would say uh giving providing

explanations about why some individuals

are behaving in such way would be

another action that actually reduces the free energy of another agent

that's very cool yeah like by providing

a narrative summary it's a sense making tool that helps

explain and make sense of their

perceptions and actions Martin

yeah it's really interesting and thanks for that uh it just got me thinking you know sometimes we

we I guess we try to free

um minimize energy and others directly like I'm a university teacher so by teaching I'm trying to reduce you know their free energy

um but sometimes you find it's best to let them

on their own way so I'm just thinking about like in fact analytics which is very distant field from my area but uh letting others interpret meaning from their situation rather than bringing it for them if that makes sense and I'm just thinking you know maybe this is a side topic but would active infants speak to what's a better approach

or want to do it for the other or to let the other explore

in a way let them work through that space of energy free energy and then find their own way to minimize that

yeah great

topics

oh yeah I think that's a really good uh point because like at in cycle analysis we sometimes like

um

have to choose whether to give a very direct interpretation or sometimes we let the patient to understand what's going on by themselves but it's always a controversial

um question about what to do

well we do have some like theories and
some sense about like when what to do on
certain situations but uh it's quite a
Pity that we don't really have like a
very uh scientific grounds for what to
do in psychoanalysis so I think that's a
really good point

let's go to mind is uh this this came up
with this chapter in fact this so if
we're working to minimize our free
energy right and we sort of
acting in a world
to minimize surprise how do we avoid
confirmatory bias

you know so I'm thinking now that the
world is flat so I am going to reinforce
that model because if I don't then
that's going to create some uh free
energy right it's going to create a
surprise so I'm gonna act in the world
I'm going to hang out with people who
don't challenge that and I'm thinking
back into psycho um

Psychiatry practice or counseling maybe
uh

people may have ingrained sort of models
in the perceiving the world just based
on that model

um

resulting confirm it's very bias

yeah this is a is a very important topic
especially as the

as they put it existential imperative is
self-evidencing

to repeatedly re-measure my body at
homeostatic temperature

to be unsurprised about re-measuring
myself at homeostatic temperature

is to maintain homeostasis
so in reward learning you would say well
homeostatic temperatures are the most
rewarding that's the top of the reward
function and then we're going to
basically Hill Climb to get to the most
rewarding body temperature
that's the reward learning paradigm
inactive your preference is your
expectations the C Factor
is what you expect expect to find myself
at body temperature that's the bottom of
the bowl I'd be the least surprised at
37c and I'd get more surprised as I went
out and now we're going to go to the
bottom of the bowl
and we're going to pursue the most
likely course of action bound our
surprise with expected and variational
free energy
and so with the imperative as
self-evidencing and surprisal bounding
we're going to pursue likely Paths of
action
rather than with the imperative being
reward learning the imperative is to
maximize reward
however
does self-evidencing mean that we
um
seek out homophily doesn't mean that we
just shut our eyes and go into the dark
room
these are really important topics and
for example in the dark room problem
um it has been discussed for many years
people say well if the imperative of the
generative model is to reduce surprise

then why don't people just go into sensory deprivation where there's highly predictable sensory inputs and this has been discussed and addressed for for a long time but the short answer is while you're in the room reducing your uncertainty about your visual field

you're increasing your uncertainty about a variety of other aspects that are required for operation of the organism and so you kind of net on balance uncertainty is rising while you're in the room

which is why we are always adaptively moving around and seeking information and all of this

however you can always set up a constrained or a toy generative model that exhibits any given limitation or pathology

that's not too surprising itself

but to actually use this as an accounting system for the kinds of adaptive and interesting behavior that organisms do exhibit that's the interesting piece not whether you can make a

um visual input only generative model that may select to be in a dark room but if you have an active visual entity that can turn on a light

it may turn on the light or if it has a hunger

uh cue it may need to seek food

eventually so it's a little bit of a

non-sequit or a kind of over

interpretation of a thought experiment

that's just really in a little bit of a

um

partial

way

trying to come at this so

you can imagine that there's situations

where self-evidencing goes right and

where it goes wrong

and that's exactly the space that we're

exploring

a lot of times when I think about

homeostasis and temperature

um I think about you know like huberman

and many uh kind of biohackers want you

to uh you know experience cold contrast

therapy and especially if you are

sensitive to cold you know it's very

important that you get in there and do

your cold plunge your cold showers and

so that's that's I think that's an

interesting example where you're

changing that you know that set point

basically

um by by doing other activities that

seem opposite to what uh you know what

would what would make you comfortable

cool thank you

okay

any other thoughts or questions or let's

look at like one or a few of these asked

questions just just a general question

Daniel so when we post questions because

I posted that one I had to go to the

main table to post a question and then

it feeds into the cohort four is that

how it works I couldn't I couldn't post

directly into there yeah

um all the uh it's it's all

there's only one underlying questions
object
so each of these are like different
views so if you just go here and just
like hit enter
you'll add a new row
and you can ask a question in cohort for
and then that'll be like in chapter one
and in cohort four
um and so that it'll come up when we go
to chapter one questions as well or you
could um
yeah or you could ask and but then it
might not necessarily be tagged to
cohort for
but it doesn't really matter what cohort
it you know as long as it's in the
chapter that's a good place got it
thanks
cool
well
um
one distinction and and you know I look
forward to seeing some of you
um
next week in the other time also
discussing chapter one
um or next week at this time where we'll
be in chapter six with cohort four
you're all welcome to join but a really
key distinction
that's a that's um raised in figure 1.2
is the high road and the low road
and this is basically going to structure
chapters two and three
the low road is going to describe how
the low road is basically Bayesian
statistics

that's how we're going to do cognitive modeling

but you could use Bayesian statistics to model anything you could do it on health records you could do it on weather data so Bayesian modeling doesn't have to be about adaptive cybernetic Auto poetic agents Bayesian statistics the how can be about anything

chapter two low wrote

chapter 3 is going to be the high road and the high road is like the kind of what or why

and it's self-organized self-evidencing systems

and so this describes more like the specifications for adaptive agents this is like what they need to be doing to be

observed repeatedly

by an observer and or by themselves in this kind of like sophisticated reflexive sense

but it doesn't necessarily say how that happens

it turns out where the why and the how meet is active inference we're going to be using Bayesian statistics that's how to model adaptive cybernetic entities that's why

and so that's kind of going to be explored in the chapters two and three low road High Road

so that's kind of the the big structuring motif ultimately of this textbook

um Olivier

yeah in

in the figure there is a an order you
know in all the items on the lower Road
and a high road uh it is although
important to understand anything or just
you know one group and another group
it's a it's a schematic ordering I
believe that there is a way to read them
in order but it's not an exhaustive or a
unique ordering so

um up

so for example

um

oh sorry someone is someone closing the
um

okay so base theorem is just describing
basically tautological statements about
conditional and Joint probabilities
this gets used in a generative model
which is basically a model that can be
used in a recognition capacity or an A
generative capacity that's the tale of
two densities

in simple settings or passive settings
we're interested in perception as
inference

but once we go into the anticipatory or
the predictive setting we're interested
not just in dealing with incoming
streaming data but basically making
ongoing predictions now casting
the next kind of into the unknown step
that now casting can take is action as
inference because it's not enough to
just deal with incoming streams of data
sometimes you want to take the action to
open the book to find out
that idea of perception and planning as
inference is Unified in the Bayesian

brain and it turns out that variational Bayesian methods are going to give us some tractable optimizable approaches for approximate Bayesian computation where exact Bays probably doesn't apply or is really challenging just like we kind of mentioned earlier so that gets us to doing variational Bayesian optimization on perception and planning as inference in the Bayesian brain that's active inference so that's kind of low road ordering the high road ordering is starting with the free energy principle which is a non-falsifiable principle just like the principle of least action in classical mechanics this is like a principle for informational systems and systems under this principle are seen as being in the game of self-organization via surprise minimization you'll continue to find yourself organized if you're continually minimizing your surprise about who and what you are that is um supported or described by a partitioning conditional independent structure called a Markov blanket which we're going to explore and the markup blanket can be seen as carrying out self-evidencing through

this kind of anticipatory real-time
predictive processing which is related
to predictive coding
and in the extended embedded inactive
setting that can be understood as
basically self-authoring self-meaning
making and ultimately even extended
Niche Construction
which gets us to active inference from
The High Road
so again not exhaust exhaustive or or
unique ordering but certainly something
that they have considered and laid out a
certain way
thank you everyone
for this awesome
um chapter one discussion
I hope to see you
um
either next week at the other time or
whenever
and just leave a comment on this page
here
if you um want to start up the math
group so thank you all
thank you
thank you